Pranam Chatterjee, PhD

One uCity Square, Room 1012, 25 N 38th St, Philadelphia, PA 19104
706-442-2715 | pranam@upenn.edu

Education

Massachusetts Institute of Technology September 2020
Ph.D. in Media Arts and Sciences
- **Thesis**: "Robust Genome Editing with Broad-Targeting CRISPR Enzymes"
- **Research:** Sequence Algorithms, Genome Editing, Protein Engineering

Massachusetts Institute of Technology June 2018
S.M. in Media Arts and Sciences
- **Thesis**: "A Novel CRISPR-Cas9 Platform with Expanded Targeting Capabilities"

Massachusetts Institute of Technology June 2016
S.B. in Computer Science and Molecular Biology
- **Thesis:** "Synthetic Meganucleases for Genome Editing"

Academic Positions

University of Pennsylvania July 2025 –
Assistant Professor of Bioengineering
Assistant Professor of Computer and Information Science
Africk-Lesley Distinguished Scholar of Innovation in Engineering
- Principal investigator of the Programmable Biology Group, developing new **discrete generative models** for the *de novo* design of biologics, spanning from algorithmic theory to pre-clinical validation *in vitro* and *in vivo*.

University of Massachusetts Medical School January 2024 –
Affiliate Assistant Professor at the RNA Therapeutics Institute

Buck Institute for Aging Research July 2022 –
Adjunct Assistant Professor

Duke University July 2022 – 2025
Assistant Professor of Biomedical Engineering
Assistant Professor of Computer Science
Assistant Professor of Biostatistics and Bioinformatics
- Principal investigator of the Programmable Biology Group, developing generative language models to design programmable proteins for applications in genome editing, proteome editing, and ovarian cell engineering.

Harvard Medical School September 2020 – June 2022
Carlos M. Varsavsky Research Fellow with Professor George Church
- Led a research team to differentiate ovarian cell types (granulosa cells and oocytes) from stem cells by developing graph theory algorithms and high-throughput genetic screening strategies for transcription factor identification

MIT Media Lab July 2016 – July 2020
Graduate Research Student with Professor Joseph Jacobson
- Independently engineered broad-targeting, efficient, and safe CRISPR enzymes, integrating computational protein design with experimental validation and optimization

Pfizer June 2014 – October 2014
Immunoscientist
- Led the computational design and experimental validation of T cell co-inhibitory receptor PD-1 agonist antibodies for immunotherapy

Harvard Medical School/Beth Israel Deaconess Medical Center June 2011 – June 2014

Undergraduate Research Student with Professor Vassiliki Boussiotis

- Led numerous projects to assess the interaction of PD-1 with downstream phosphatases, determined the metabolic profile of CD4+ T cells downstream of PD-1 signaling, and ascertained the role of various cytokines in thymopoietic reconstitution after transplantation

Dartmouth Hitchcock Medical Center January 2012 – May 2013

Undergraduate Research Student with Professor Mark Israel

- Studied the modulation of mitochondrial biogenesis by ID2 in glioma

Fort Valley State University June 2010 – May 2011

High School Research Student with Professor Nirmal Joshee

- Conducted individual research on *Scutellaria* genus plants, performing micropropagation and genetic transformation protocols

Peer-Reviewed Publications and Proceedings

1. Peng FZ, Bezemek Z, Patel S, Rector-Brooks J, Yao S, Bose J, **Chatterjee P**†, Tong A†. Path Planning for Diffusion Language Model Sampling. *Transactions on Machine Learning Research (TMLR)*, 2026.
2. Cao H, Pal A, Kimura S, Kim Y, Zhang J, Heng PA, **Chatterjee P**†. AlloGen: Conformation-Selective Binder Design with Differential State Scoring. *Conference on Neural Information Processing Systems (NeurIPS)*, 2026.
3. Tang S, Zhang Y, **Chatterjee P**†. Entangled Schrödinger Bridge Matching. *Conference on Neural Information Processing Systems (NeurIPS)*, 2026.
4. Zhang R, Li Z, Gu J, **Chatterjee P**†. CupOFMoCA: Coupled Objective-Guided Discrete Flows for Molecular Conjugate Assembly. *Conference on Neural Information Processing Systems (NeurIPS)*, 2026.
5. Chen T, Holsman M, Zhao L, Zhang Y, **Chatterjee P**†. pCoMole: Pareto-Constrained Molecule Editing with Discrete Flows. *Conference on Neural Information Processing Systems (NeurIPS)*, 2026.
6. Tang S, **Chatterjee P**†. Expanding Flow Maps. *Conference on Neural Information Processing Systems (NeurIPS)*, 2026.
7. De Santi R, Lee BD, Perez Jensen C, Protopapas K, Tang S, Liu CH, **Chatterjee P**, Yue Y, Krause A. Active Flow Expansion for Out-of-Distribution Discovery: from Theory to Molecules. *Conference on Neural Information Processing Systems (NeurIPS)*, 2026.
8. Ye T, Alamgir A, Robertus CM, Colina D, Monticello C, Donahue TC, Bandi C, Hong L, Vincoff S, Goel S, Lam K, Heyes J, Putnam D, Alabi CA, **Chatterjee P**†, DeLisa MP†. Programmable protein degraders enable selective knockdown of pathogenic β-catenin subpopulations in vitro and in vivo. *Science Advances*, 12(35): eaec8271 (2026).
9. Wan S, Gold J, Vure P, Mogilevsky CS, Talikoti A, Chen T, Gupta A, Biswas T, You Z, Acharya V, **Chatterjee P**, Wang X, Gao X. Adaptive model-guided protein evolution with sparse data optimizes compact eukaryotic genome editors. *Nature Biotechnology*, 2026.
10. Cao H, Pal A, Tang S, Zhang Y, Zhang J, Heng PA, **Chatterjee P†.** TD3B: Transition-Directed Discrete Diffusion for Allosteric Binder Generation. *International Conference of Machine Learning*, 2026 (Spotlight).
11. Zhang Y, Tang S, Chen T, Mahood E, Vincoff S, **Chatterjee P†.** PeptiVerse: A Unified Platform for Therapeutic Peptide Property Prediction. *Nature Communications*, 17(1): 6819 (2026).
12. Hong L, Vincoff S, **Chatterjee P**†. AI-Designed Peptides as Tools for Biochemistry. *Biochemistry*, 65(8): 1071–1083 (2026).
13. Tang S, Zhang Y, Tong A, **Chatterjee P**†. Branched Schrödinger Bridge Matching. *International Conference of Learning Representations*, 2026.
14. Tsurumi A, Cahill CM, Liu AJ, **Chatterjee P**, Das S, Kobayashi A. Development of plasma protein classification models for Alzheimer's disease using multiple machine learning approaches. *Int J Mol Sci*. 2025;26(23):11673, 2025.
15. Torres MDT, Chen T, Wan F, **Chatterjee P**†, de la Fuente-Nunez, C†. Generative latent diffusion language modeling yields anti-infective synthetic peptides. *Cell Biomaterials*, 1(9): 100183 (2025).
16. Chen T, Quinn Z, Dumas M, Peng C, Hong L, Lopez-Gonzalez M, Mestre A, Watson R, Vincoff S, Zhao L, Wu J, Stavrand A, Schaepers-Cheu M, Wang TZ, Srijay D, Monticello C, Vure P, Pulugurta R, Pertsemlidis S, Kholina K, Goel S, DeLisa MP, Chi JTA, Truant R, Aguilar HC, **Chatterjee P†.** Target sequence-conditioned design of peptide binders using masked language modeling. *Nature Biotechnology*, 44(6): 1002–1010 (2026).
17. Tang S, Zhang Y, **Chatterjee P**†. PepTune: De Novo Generation of Therapeutic Peptides with Multi-Objective-Guided Discrete Diffusion. *International Conference on Machine Learning*, PMLR 267: 59017–59065 (2025).
18. Hong L, Ye T, Wang TZ, Srijay D, Liu H, Zhao L, Watson R, Vincoff S, Chen T, Kholina K, Goel S, DeLisa MP, **Chatterjee P**†. Programmable Protein Stabilization with Language Model-Derived Peptide Guides. *Nature Communications*, 16(1): 3555 (2025).

19. Peng Z, Wang C, Chen T, Vincoff S, **Chatterjee P**†. PTM-Mamba: A PTM-Aware Protein Language Model with Bidirectional Gated Mamba Blocks. *Nature Methods*, 22(5): 945–949 (2025).
20. Vincoff S, Goel S, Kholina K, Pulugurta R, Vure P, **Chatterjee P**†. FusOn-pLM: A Fusion Oncoprotein-Specific Language Model via Adjusted Rate Masking. *Nature Communications*, 16(1): 1436 (2025).
21. Rector-Brooks J, Hasan M, Peng Z, Quinn Z, Liu C, Mittal S, Dziri N, Bronstein M, Bengio Y, **Chatterjee P**, Tong A, Bose AJ. Steering Masked Discrete Diffusion Models via Discrete Denoising Posterior Prediction. *International Conference of Learning Representations*, 2025.
22. Smela MP, Kramme C, Fortuna P, Wolf B, Goel S, Adams J, Ma Z, Velychko S, Widocki U, Kavirayuni VS, Chen T, Vincoff S, Dong E, Kohman R, Kobayashi M, Shioda T, Church G, **Chatterjee P**. Rapid Human Oogonia-like Cell Specification via Transcription Factor-Directed Differentiation. *EMBO Reports*, 26(4): 1114–1143 (2025).
23. Bhat S, Palepu K, Hong L, Mao J, Ye T, Iyer R, Zhao L, Chen T, Vincoff S, Watson R, Wang T, Srijay D, Kavirayuni VS, Kholina K, Goel S, Vure P, Deshpande AH, Soderling S, DeLisa M, **Chatterjee P**†**.** *De Novo* Design of Peptide Binders to Conformationally Diverse Targets with Contrastive Language Modeling. *Science Advances*, 11(4): eadr8638 (2025).
24. Chen T, **Chatterjee P**†**.** Synergizing sequence and structure representations to predict protein variants. *Cell Research*, 34(9): 597–598 (2024).
25. Paulsen B, Piechota S, Barrachina F, Giovannini A, Kats S, Potts KS, Rockwell G, Marchante M, Estevez SL, Noblett AD, Figueroa AB, Aschenberger C, Kelk DA, Forti M, Marcinyshyn S, Wiemer K, Sanchez M, Belchin P, Lee JA, Buyuk E, Slifkin RE, Pierson Smela M, Fortuna PRJ, **Chatterjee P**, McCulloh DH, Copperman AB, Ordonez-Perez D, Klein JU, Kramme CC. Rescue in vitro maturation using ovarian support cells of human oocytes from conventional stimulation cycles yields oocytes with improved nuclear maturation and transcriptomic resemblance to in vivo matured oocytes. *Journal of Assisted Reproduction and Genetics*, 41(8): 2021–2036 (2024).
26. Piechota S, Marchante M, Giovannini A, Paulsen B, Potts KS, Rockwell G, Aschenberger C, Noblett AD, Figueroa AB, Sanchez M, Barrachina F, Wiemer K, Guzman L, Belchin P, Smela MP, Fortuna PRJ, **Chatterjee P**, Tran ND, Kelk DA, Forti M, Marcinyshyn S, Smith T, McCulloh DH, Fernandez-Gonzalez MJ, Abittan B, Ortiz S, Klein JU, Klatsky P, Ordonez-Perez D, Kramme CC. Human-induced pluripotent stem cell-derived ovarian support cell co-culture improves oocyte maturation in vitro after abbreviated gonadotropin stimulation. *Human Reproduction*, 38(12), 2456-2469 (2023).
27. Brixi G, Ye T, Hong L, Wang T, Monticello C, Lopez-Barbosa N, Vincoff S, Yudistyra V, Zhao L, Haarer E, Chen Pertsemlidis S, Palepu K, Bhat S, Christopher J, Li X, Liu T, Zhang S, Petersen L, DeLisa M, **Chatterjee P**†. SaLT&PepPr is an interface-predicting language model for designing peptide-guided protein degraders. *Communications Biology*, 6(1): 1081 (2023).
28. Zhao L, Koseki S, Silverstein RA, Amrani N, Peng C, Kramme C, Savic N, Pacesa M, Rodriguez T, Stan T, Tysinger E, Hong L, Yudistyra V, Ponnapati M, Jacobson JM, Jakimo N, Church GM, Truant R, Jinek M, Kleinstiver BP, Sontheimer EJ, **Chatterjee P**†**.** PAM-Flexible Genome Editing with an Engineered Chimeric Cas9. *Nature Communications*, 14(1): 6175 (2023).
29. Chen T, Hong L, Yudistyra V, Vincoff S, **Chatterjee P**†. Generative Design of Therapeutics that Bind and Modulate Protein States. *Current Opinion in Biomedical Engineering*, 28: 100496 (2023).
30. Liang SQ, Liu P, Ponnienselvan K, Suresh S, Chen Z, Kramme C, **Chatterjee P**, Zhu LJ, Sontheimer EJ, Xue W, Wolfe S. Genome-wide Profiling of Prime editor off-target sites in vitro and in vivo using PE-tag. *Nature Methods*, 20(6): 898–907 (2023).
31. Smela M, Kramme C, Fortuna P, Adams J, Su R, Dong E, Kobayashi M, Brixi G, Kavirayuni V, Tysinger E, Kohman R, Shioda T, **Chatterjee P**, Church G. Directed differentiation of human iPSCs to functional ovarian granulosa-like cells via transcription factor overexpression. *eLife* 12:e83291 (2023).
32. Tripathy S, Ponnapati M, Bhat S, Jacobson JM, **Chatterjee P**†. Femtomolar detection of SARS-CoV-2 via peptide beacons integrated on a miniaturized TIRF microscope. *Science Advances*, 8(34): eabn2378 (2022).
33. Kobayashi M, Kobayashi M, Odajima J, Shioda K, Hwang YS, Sasaki K, **Chatterjee P**, Kramme C, Kohman R, Church GM, Loehr AM, Weiss RS, Juppner H, Gell J, Lau C, Shioda T. Expanding homogeneous culture of human primordial germ cell-like cells maintaining germline features without serum or feeder layers. *Stem Cell Reports*, 17(3): 507–521 (2022).
34. Kramme C, Plesa AM, Wang HH, Wolf B, Smela MP, Guo X, Kohman R, **Chatterjee P**†, Church GM. MegaGate: A toxin-less gateway molecular cloning tool. *STAR Protocols*, 2(4): 100907 (2021).
35. Kramme C, Plesa AM, Wang HH, Wolf B, Smela MP, Guo X, Kohman R, **Chatterjee P**†, Church GM. STAMP-Screen: An Integrated Pipeline for Mammalian Genetic Screening. *Cell Reports Methods*, 1(6): 100082 (2021).
36. **Chatterjee P**†**.** BARBEKO'ing in the lab: Versatile CRISPR screens with barcoded base editors. *Molecular Cell*, 81, 15: 3046-3047 (2021).
37. **Chatterjee P**†**,** Ponnapati M, Kramme C, Plesa AM, Church GM, Jacobson JM. Targeted intracellular degradation of SARS-CoV-2 via computationally optimized peptide fusions. *Communications Biology*, 3(1): 715 (2020).

38. Qing R*****, Tao F*****, **Chatterjee P***, Yang G, Han Q, Chung H, Ni J, Suter B, Kubicek J, Maertens B, Schubert T, Blackburn C, Zhang S. Non-full-length water-soluble $CXCR4^{QTY}$ and $CCR5^{QTY}$ chemokine receptors: implication for overlooked truncated but functional membrane receptors. *iScience*, 23(12): 101670 (2020).
39. **Chatterjee P†,** Lee J, Nip L, Koseki S, Tysinger E, Sontheimer EJ, Jacobson JM, Jakimo N. A Cas9 with PAM Recognition for Adenine Dinucleotides. *Nature Communications*, 11(1): 2474 (2020).
40. **Chatterjee P†**, Jakimo N, Lee J, Amrani N, Rodriguez T, Koseki S, Tysinger E, Qing R, Hao S, Sontheimer EJ, Jacobson JM. An engineered ScCas9 with broad PAM range and high specificity and activity. *Nature Biotechnology,* 38(10):1154-1158 (2020).
41. **Chatterjee P†**, Jakimo N, Jacobson JM. Minimal PAM specificity of a highly similar SpCas9 ortholog. *Science Advances*, 4, 10: eaau0766 (2018).
42. Zhang Z, Rahme G, **Chatterjee P,** Havrda M, Israel M. ID2 promotes survival of glioblastoma cells during metabolic stress by regulating mitochondrial function. *Cell Death and Disease*, 8(2): e2615 (2017).
43. Patsoukis N, Bardhan K, **Chatterjee P**, Sari D, Liu B, Bell L, Karoly E, Freeman G, Petkova V, Seth P, Li L, Boussiotis VA. PD-1 alters T cell metabolic reprogramming by inhibiting glycolysis and promoting lipolysis and fatty acid oxidation. *Nature Communications*, 6(1): 6692 (2015).
44. Boussiotis VA, **Chatterjee P**, Li L. Biochemical Signaling of PD-1 on T Cells and Its Functional Implications. *The Cancer Journal*, 20, 4: 265-71 (2014).

Peer-Reviewed, Non-Archival Proceedings

1. Maganti R, Maganti R, Alameh MG, **Chatterjee P†**. COMPOSE: Generative Stochastic Rewriting for Trans-Dimensional Molecular Design and Control. *NeurIPS Workshop on AI for Stochastic Dynamics: From Theoretical Foundations to Scientific Applications (STODY)*, December 2026.
    - *Also accepted to the NeurIPS Workshop on AI for Drug Discovery: Bridging the Translation Gap (AI4DD)*
2. Zhang J, Cao H, Zhang Y, Wang Y, Gu C, Heng PA, **Chatterjee P†**. StapleBridge: Chemistry-Aware Schrödinger Control for Hydrocarbon-Stapled Peptide Lead Optimization. *NeurIPS Workshop on AI for Drug Discovery: Bridging the Translation Gap (AI4DD)*, December 2026.
3. Nori N, Mahood EH, Liang K, **Chatterjee P†**. TreeSBM: Tree-Valued Schrödinger Bridges for Protein Evolution Forecasting. *NeurIPS Workshop on AI for Stochastic Dynamics: From Theoretical Foundations to Scientific Applications (STODY)*, December 2026.
    - *Also accepted to the NeurIPS Workshop on Generative and Experimental Perspectives for Biomolecular Design (GEM)*
4. Zhao L, Mattix AA, Pal A, Chen T, Vincoff S, Hong L, Renteria DC, Sase S, Vanderver A, Matson DR, **Chatterjee P†**. A Mammalian High-Throughput Screen for AI-Designed Peptide-Guided Protein Degraders. *NeurIPS Workshop on Generative and Experimental Perspectives for Biomolecular Design (GEM)*, December 2026.
5. Zhang R, Mahood EH, Gu J, **Chatterjee P†**. LOOPY: Guiding Generative Design with Locally Calibrated Predictor Uncertainty. *NeurIPS Workshop on AI for Drug Discovery: Bridging the Translation Gap (AI4DD)*, December 2026.
    - *Also accepted to the NeurIPS Workshop on Generative and Experimental Perspectives for Biomolecular Design (GEM)*
6. Rahman F, **Chatterjee P†**. Nonparametric Biologic Design via Discrete Drifting. *NeurIPS Workshop on Generative and Experimental Perspectives for Biomolecular Design (GEM)*, December 2026.
7. Mahood EH, Zhang Y, Taylor MK, **Chatterjee P†**. DiFfER: Discrete Flows for Enzymatic Reaction Mechanism Step Prediction. *NeurIPS Workshop on Machine Learning for Simulations in Biology and Chemistry (SIMBIOCHEM)*, December 2026.
    - *Also accepted to the NeurIPS Workshop on Generative and Experimental Perspectives for Biomolecular Design (GEM)*
8. Maganti R, Maganti R, Alameh MG, **Chatterjee P†**. FORGE: Reaction-Guided Generative Design of Ionizable Lipids. *NeurIPS Workshop on Generative and Experimental Perspectives for Biomolecular Design (GEM)*, December 2026.
9. Cao H, Heng PA, **Chatterjee P†**. AlloShift: Population-Directed Binder Design with State-Selective Molecular Landscapes. *NeurIPS Workshop on Machine Learning for Simulations in Biology and Chemistry (SIMBIOCHEM)*, December 2026.
10. Mahood EH, Chatterjee P†. SPROUT: Steered Plant Promoter Editing via Rollout-Guided Utility Tilting of Edit Flows. *ICML Workshop on Generative and Agentic AI for Biology*, July 2026.
11. Vincoff S, Chatterjee P†. SOAPIA: Specificity-Guided Generation of Off-Target-Avoiding Protein Interactions with High Target Affinity. *ICML Workshop on Generative and Agentic AI for Biology*, July 2026.
12. Liang K, Nori N, Mahood EH, Zhang Y, Chatterjee P†. PROPHET: Phylogenetically Robust Antiviral Peptide Design Against Heterogeneous Evolutionary Trajectories. *ICML Workshop on Generative and Agentic AI for Biology*, July 2026.

13. Chen T, Holsman M, Chatterjee P†. PIGEON: Pocket-Inferred Geometric Ensemble Flexible Docking. *ICML Workshop on Generative and Agentic AI for Biology*, July 2026.
14. Li Z, Zhang R, Chatterjee P†. CupOFLATTE: Coupled Objective-Guided Discrete Flows via Linker Assembly for Targeted PROTAC Engineering. *ICML Workshop on Generative and Agentic AI for Biology*, July 2026.
15. Cao H, Pal A, Kimura S, Kim Y, Zhang J, Heng PA, Chatterjee P†. AlloGen: Conformation-Selective Binder Design with Differential State Scoring. *ICML Workshop on Generative and Agentic AI for Biology*, July 2026.
16. Cao H, Gao Z, Gu C, Liu G, Heng PA, **Chatterjee P†.** SF-Cluster: Frustration-Aware MSA Subsampling for Protein Conformational Control. *ICML AI for Science Workshop (Oral)*, July 2026.
17. De Santi R, Lee BD, Perez Jensen C, Protopapas K, Tang S, Liu CH, **Chatterjee P**, Yue Y, Krause A. Active Flow Expansion for Out-of-Distribution Discovery: from Theory to Molecules. *International Conference on Machine Learning Workshop on Structured Probabilistic Inference & Generative Modeling*, July 2026.
    - *Also accepted to ICML Workshops: Workshop on Generative and Agentic AI for Biology*
18. Tang S, **Chatterjee P†.** Expanding Flow Maps. *ICML Workshop on Structured Probabilistic Inference & Generative Modeling*, July 2026.
19. Liang K, Chen T, **Chatterjee P†.** LaPep: Can Language Contribute to Property-Guided Peptide Design? *ICLR Workshop on Logical Reasoning of Large Language Models*, April 2026.
20. Zhang R, Gu J, **Chatterjee P†.** CupOFMoCA: Coupled Objective-Guided Discrete Flows for Molecular Conjugate Assembly. *ICLR Workshop on Real-World Constrained and Preference-Aligned Flow and Diffusion-Based Models*, April 2026.
    - *Also accepted to ICLR Workshops: Deep Generative Model in Machine Learning: Theory, Principle and Efficacy*
21. Mahood E, Pattie W, Jones EM, Vincoff S, Masiach A, **Chatterjee P†.** MOOSE: Targeted Small Molecule Generation via Multi-Objective-Guided Discrete Diffusion. *ICLR Workshop on Real-World Constrained and Preference-Aligned Flow and Diffusion-Based Models*, April 2026.
    - *Also accepted to ICLR Workshops: Foundation Models for Science*
22. Hong L, Ho TC, Tseng YS, Vincoff S, Chen T, Chen P, **Chatterjee P†.** PepRePs: Peptide-Retargeted Phosphatases via Generative Language Models. *ICLR Workshop on Learning Meaningful Representations of Life*, April 2026.
    - *Also accepted to ICLR Workshops: Foundation Models for Science*
23. Zhang Y, Tang S, Chen T, Mahood E, Vincoff S, **Chatterjee P†.** PeptiVerse: A Unified Platform for Therapeutic Peptide Property Prediction. *ICLR Workshop on Generative and Experimental Perspectives for Biomolecular Design*, April 2026.
24. Vincoff S, **Chatterjee P†.** SNOOPPI: Sequence-Normalized Database of On- and Off-Target Protein-Protein Interactions. *ICLR Workshop on Learning Meaningful Representations of Life*, April 2026.
    - *Also accepted to ICLR Workshops: Generative and Experimental Perspectives for Biomolecular Design* and *Foundation Models for Science*
25. Goel S, **Chatterjee P†.** Minimal-Action Discrete Schrödinger Bridge Matching for Peptide Sequence Design. *ICLR Workshop on Real-World Constrained and Preference-Aligned Flow and Diffusion-Based Models*, April 2026.
    - *Also accepted to ICLR Workshops: Deep Generative Model in Machine Learning: Theory, Principle and Efficacy*
26. Patel S, Tang S, **Chatterjee P**†, Yao S†. Full-length mRNA Design with Reward-Guided Masked Diffusion Model Fine-Tuning. *ICLR Workshop on Real-World Constrained and Preference-Aligned Flow and Diffusion-Based Models*, April 2026.
    - *Also accepted to ICLR Workshops: Foundation Models for Science*
27. Cao H, Pal A, Tang S, Zhang Y, Zhang J, Heng PA, **Chatterjee P†.** TD3B: Transition-Directed Discrete Diffusion for Allosteric Binder Generation. *ICLR Workshop on Real-World Constrained and Preference-Aligned Flow and Diffusion-Based Models*, April 2026.
    - *Also accepted to ICLR Workshops: Generative and Experimental Perspectives for Biomolecular Design, Learning Meaningful Representations of Life*, *Deep Generative Model in Machine Learning: Theory, Principle and Efficacy*
28. Tang S, Zhu Y, Tao M, **Chatterjee P†.** A2D2: Finetuning Any-Length Discrete Diffusion for Adaptive Decoding. *ICLR Workshop on Real-World Constrained and Preference-Aligned Flow and Diffusion-Based Models*, April 2026.
29. Ponnapati M, Chen T, Zhang Y, **Chatterjee P†.** Controlling Target Protein Dynamics via Lie-Guided Discrete Flows. *ICLR Workshop on Real-World Constrained and Preference-Aligned Flow and Diffusion-Based Models*, April 2026.
    - *Also accepted to ICLR Workshops: Generative and Experimental Perspectives for Biomolecular Design*

30. Chen T, Holsman M, Zhao L, Zhang Y, **Chatterjee P†.** pCoMole: Pareto-Constrained Molecule Editing with Discrete Flows. *ICLR Workshop on Real-World Constrained and Preference-Aligned Flow and Diffusion-Based Models*, April 2026.
    - *Also accepted to ICLR Workshops: Generative and Experimental Perspectives for Biomolecular Design, Learning Meaningful Representations of Life*, *Deep Generative Model in Machine Learning: Theory, Principle and Efficacy*
31. Chen T, Zhang Y, Vincoff S, **Chatterjee P†.** AReUReDi: Annealed Rectified Updates for Refining Discrete Flows with Multi-Objective Guidance. *ICLR Workshop on Real-World Constrained and Preference-Aligned Flow and Diffusion-Based Models*, April 2026.
    - *Also accepted to ICLR Workshops: Generative and Experimental Perspectives for Biomolecular Design, Learning Meaningful Representations of Life*, *Deep Generative Model in Machine Learning: Theory, Principle and Efficacy*
32. Goel S, Schray PM, Zhang Y, Vincoff S, Kratochvil H, **Chatterjee P†.** Token-Level Guided Discrete Diffusion for Membrane Protein Design. *NeurIPS Workshop on AI for Science*, December 2025.
    - *Also accepted to NeurIPS Workshops: Structured Probabilistic Inference & Generative Modeling,* and *Multi-modal Foundation Models and Large Language Models for Life Sciences, Frontiers in Probabilistic Inference*)
33. Chen T, Quinn Z, Zhang Y, **Chatterjee P†.** moPPIt-v3: Motif-Specific Peptides Generated via Multi-Objective-Guided Discrete Flow Matching. *NeurIPS Workshop on Frontiers in Probabilistic Inference,* December 2025.
    - *Also accepted to NeurIPS Workshops: AI for Science, Structured Probabilistic Inference & Generative Modeling*
34. Zhang Y, Tang S, **Chatterjee P†.** ScooBDoob: Schrödinger Bridge with Doob's h-Transform for Molecular Dynamics. *NeurIPS Workshop on Frontiers in Probabilistic Inference,* December 2025.
    - *Also accepted to NeurIPS Workshops: Structured Probabilistic Inference & Generative Modeling*
    - *Accepted to EurIPS Workshop: Machine Learning for Simulations in Biology and Chemistry* (Spotlight Award)
35. Tang S, Zhang Y, **Chatterjee P†.** Entangled Schrödinger Bridge Matching. *NeurIPS Workshop on Frontiers in Probabilistic Inference,* December 2025.
    - *Also accepted to NeurIPS Workshops: Structured Probabilistic Inference & Generative Modeling*
36. Zhang R, **Chatterjee P†.** Multi-Objective Nanobody Design via Masked Discrete Diffusion with Simplex Refinement. *NeurIPS Workshop on Frontiers in Probabilistic Inference*, December 2025.
    - *Also accepted to NeurIPS Workshops: AI for Science, Structured Probabilistic Inference & Generative Modeling*
37. Zhao L, Pal A, Chen T, **Chatterjee P†.** High-Throughput Protein Perturbation Screens with AI-Designed Degraders. *NeurIPS Workshop on AI Virtual Cells and Instruments*, December 2025.
38. Tang S, Zhang Y, Tong A, **Chatterjee P†**. Branched Schrödinger Bridge Matching. *ICML Workshop on Exploration in AI Today*, July 2025.
39. Chen T, Zhang Y, Tang, S, **Chatterjee P†**. Multi-Objective-Guided Discrete Flow Matching for Controllable Biological Sequence Design. *ICML Workshop on Generative Biology (GenBio)*, July 2025. *Spotlight Award.*
    - *Also accepted to ICML Workshops: Generative Biology* and *Multi-modal Foundation Models and Large Language Models for Life Sciences*
40. Vincoff S, Davis O, Ceylan II, Tong A, Bose J, **Chatterjee P†**. SOAPIA: Siamese-Guided Generation of Off Target-Avoiding Protein Interactions with High Target Affinity. *ICML Workshop on Multi-modal Foundation Models and Large Language Models for Life Sciences*, July 2025.
    - *Also accepted to ICML Workshops: Generative Biology* and *Scaling Up Intervention Models (SIM)*
41. Patel S, Tang S, Zhang Y, **Chatterjee P†**, Yao S†. Multi-Objective-Guided Generative Design of mRNA with Therapeutic Properties. *ICML Generative Biology Workshop*, July 2025.
    - *Also accepted to ICML Workshops: Multi-modal Foundation Models and Large Language Models for Life Sciences* and *Scaling Up Intervention Models (SIM)*
42. Tang S, Zhang Y, **Chatterjee P†**. PepTune: De Novo Generation of Therapeutic Peptides with Multi-Objective-Guided Discrete Diffusion. *ICLR Workshop on Frontiers in Probabilistic Inference*, March 2025.
    - *Also accepted to ICLR Workshops: Generative and Experimental Perspectives for Biomolecular Design* and *Deep Generative Models in Machine Learning*
43. Peng FZ, Bezemek Z, Patel S, Rector-Brooks J, Yao S, Tong A, **Chatterjee P†**. Path Planning for Masked Diffusion Models with Applications to Biological Sequence Generation. *ICLR Workshop on Deep Generative Models in Machine Learning*, March 2025 (Best Paper Award).
    - *Also accepted to ICLR Workshops: Generative and Experimental Perspectives for Biomolecular Design* and *Frontiers in Probabilistic Inference*

44. Zhang Y, Srijay D, **Chatterjee P†.** Metalorian: De Novo Generation of Heavy Metal-Binding Peptides with Classifier-Guided Diffusion Sampling. *ICLR Workshop on Learning Meaningful Representations of Life*, March 2025.
    - *Also accepted to ICLR Workshop: Generative and Experimental Perspectives for Biomolecular Design*
45. Chen T, Zhang Y, **Chatterjee P†**. moPPIt: De Novo Generation of Motif-Specific Peptide Binders via Conditional Uniform Discrete Diffusion. *ICLR Workshop on Learning Meaningful Representations of Life*, March 2025.
    - *Also accepted to ICLR Workshop: Generative and Experimental Perspectives for Biomolecular Design*
46. Goel S, Thoutam V, Marroquin EM, Gokaslan A, Firouzbakht A, Vincoff S, Kuleshov V, Kratochvil HT, **Chatterjee P†**. MeMDLM: De Novo Membrane Protein Design with Property-Guided Discrete Diffusion. *ICLR Workshop on Learning Meaningful Representations of Life*, March 2025.
    - *Also accepted to ICLR Workshop: Generative and Experimental Perspectives for Biomolecular Design*
47. Zhao L, Pal A, Chen T, **Chatterjee P†**. A Mammalian High-Throughput Assay to Screen AI-Designed Protein Degraders. *ICLR Workshop on Generative and Experimental Perspectives for Biomolecular Design*, March 2025.
48. Watson R, Patel K, **Chatterjee P†**. Targeting Aggregating Proteins with Language Model-Designed Degraders. *ICLR Workshop on Generative and Experimental Perspectives for Biomolecular Design*, March 2025.
49. Vincoff S, Davis O, Tong A, Bose J, **Chatterjee P†**. SOAPI: Siamese-Guided Generation of Off-Target-Avoiding Protein Interactions. *ICLR Workshop on Learning Meaningful Representations of Life*, March 2025.
    - *Also accepted to ICLR Workshop: Generative and Experimental Perspectives for Biomolecular Design*
50. Tang S, Zhang Y, Tong A, **Chatterjee P†**. Gumbel-Softmax Score and Flow Matching for Discrete Biological Sequence Generation. *ICLR Workshop on Frontiers in Probabilistic Inference*, March 2025.
    - *Also accepted to ICLR Workshop: AI for Nucleic Acids*
51. Chen T, **Chatterjee P†**. muPPIt: De Novo Generation of Mutant-Specific Peptide Binders via Conditional Uniform Discrete Diffusion. *ICLR Workshop on Learning Meaningful Representations of Life*, March 2025.
52. Patel S, Peng FZ, Fraser K, Friedman AD, **Chatterjee P†**, Yao S. EvoFlow-RNA: Generating and Representing Non-Coding RNA with a Language Model. *ICLR Workshop on AI for Nucleic Acids*, March 2025.
53. Wang L, Pulugurta R, Vure P, Pal A, Zhang Y, **Chatterjee P†**. PepDoRA: A Unified Peptide Language Model via Weight-Decomposed Low-Rank Adaptation. *NeurIPS AIDrugX Workshop*, December 2024.
54. Goel S, Thoutam V, Marroquin EM, Gokaslan A, Firouzbakht A, Vincoff S, Kuleshov V, Kratochvil HT, **Chatterjee P†**. MeMDLM: De Novo Membrane Protein Design with Masked Discrete Diffusion Protein Language Models. *NeurIPS AIDrugX Workshop*, December 2024.
55. Patel S, Fraser K, Peng Z, Friedman AD, Yao O, **Chatterjee P†**, Yao S. AptaBLE: A Deep Learning Platform for SELEX Optimization. *NeurIPS AIDrugX Workshop*, December 2024.
56. Chen T, Zhang Y, **Chatterjee P†.** moPPIt: *De Novo* Generation of Motif-Specific Binders with Protein Language Models. *NeurIPS Machine Learning in Structural Biology Workshop*, December 2024.
57. Zhang Y, He P, Hsu A, **Chatterjee P†**. MetaLATTE: Metal Binding Prediction via Contrastive Learning on Protein Language Model Latents. *ICML 2024 Workshop on Machine Learning for Life and Materials Science*, July 2024.
58. Peng Z, Schussheim B, **Chatterjee P†**. PTM-Mamba: A PTM-Aware Protein Language Model with Bidirectional Gated Mamba Blocks. *ICML 2024 Workshop on Machine Learning for Life and Materials Science*, July 2024.
59. Vincoff S, Goel S, Kholina K, **Chatterjee P†**. FusOn-pLM: A Fusion Oncoprotein-Specific Language Model via Focused Probabilistic Masking. *ICML 2024 Workshop on Efficient and Accessible Foundation Models for Biological Discovery*, July 2024.
60. Zhang Y, He P, Hsu A, Vincoff S, **Chatterjee P†**. PPI-Llama2: De Novo Generation of Binding Proteins Conditioned On Target Sequence Alone. *ICLR Workshop on Generative and Experimental Perspectives for Biomolecular Design*, May 2024.
61. Hong L, Ye T, Wang TZ, Srijay D, Zhao L, Vincoff S, Chen T, DeLisa MD, **Chatterjee P†**. Programmable Protein Stabilization with Language Model-Derived Peptide Guides. *ICLR Workshop on Generative and Experimental Perspectives for Biomolecular Design*, May 2024-2025.
62. Zhao L, Mohan A, Patel A, **Chatterjee P†**. A High-Throughput Human Display Screen to Identify Target-Specific Binder Proteins via Chimeric Antigen Receptors. *ICLR Workshop on Generative and Experimental Perspectives for Biomolecular Design*, May 2024.
63. Chen T, Pertsemlidis S, **Chatterjee P†**. PepMLM: Target Sequence-Conditioned Generation of Peptide Binders via Masked Language Modeling. *ICLR Workshop on Generative and Experimental Perspectives for Biomolecular Design*, May 2024.
64. Chen T, Vure P, Pulugurta R, **Chatterjee P†**. AMP-Diffusion: Integrating Latent Diffusion with Protein Language Models for Antimicrobial Peptide Generation. *NeurIPS Workshop on Generative AI and Biology*, October 2023.
65. Bardhan K, Patsoukis N, Sari D, Anagnostou T, **Chatterjee P**, Freeman G, Li L, Boussiotis VA. PD-1 Inhibits TCR Proximal Signaling By Sequestering SHP-2 Phosphatase and Facilitating CSK-Mediated Inhibitory Phosphorylation of Lck. *Blood*, 126(23): 283 (2015).
66. Patsoukis N, **Chatterjee P**, Sari D, Petkova V, Li L, Boussiotis VA. PD-1 Induces Metabolic Reprogramming of Activated T Cells from Glycolysis to Lipid Oxidation. *Blood*, 122(21): 187 (2013).

67. **Chatterjee P**, Patsoukis N, Boussiotis VA. Distinct Roles of PD-1 ITSM and ITIM In Regulating Interactions With SHP-2, ZAP-70, and Lck, and PD-1-Mediated Inhibitory Function. *Blood*, 122(21): 191 (2013).
68. **Chatterjee P**, Patsoukis N, Sari D, Boussiotis VA. PD-1 Couples Glucose Starvation and Survival Through AMPK-mediated Phosphorylation of Ulk1. *Blood*, 120(21): 836 (2012).

Preprints

1. De Santi R, Di Gennaro F, Gnazzo A, Tang S, **Chatterjee P**, Yang F, Ceylan II, Bronstein MM, Krause A. LLM-to-Flow Distillation: Teaching Natural Language Goals to Scientific Generators. *arXiv*, October 2026.
2. Batzolis G, Kontos G, Trentini B, **Chatterjee P**, Livne M, Girolami M, Ambrogioni L. Multimodal Co-Generation With Continuous Bitstream Diffusion. *arXiv*, October 2026.
3. Zhao L, Mattix A, Pal A, Chen T, Vincoff S, Hong L, Renteria D, Sase S, Vanderver AL, Matson DR, **Chatterjee P**†. A Mammalian High-Throughput Screen for AI-Designed Peptide-Guided Protein Degraders. *bioRxiv*, August 2026.
4. De Santi R, Lee BD, Perez Jensen C, Protopapas K, Tang S, Liu CH, **Chatterjee P**, Yue Y, Krause A. Active Flow Expansion for Out-of-Distribution Discovery: from Theory to Molecules. *arXiv*, June 2026.
5. Patel S, Tang S, Kim Y, Zhang Y, Srijay D, Lin PJ, Shubham S, Pi F, Wu C, Yao S, **Chatterjee P**†. mRNAutilus: Multi-Objective-Guided Discrete Generation of mRNA with Optimized Therapeutic Properties. *arXiv*, May 2026.
6. Zhang J, Cao H, Shi H, Mutian H, Wang Y, Gao Z, Wu F, Yao X, Hsieh CY, Pan SJ, **Chatterjee P**, Gu C, Heng PA. GeoCycler: Reward-Aligned 3D Diffusion for Constraint-Conditioned Cyclic Peptide Design. *arXiv*, May 2026.
7. Goel S, **Chatterjee P**†. Minimal-Action Discrete Schrödinger Bridge Matching for Peptide Sequence Design. *arXiv*, February 2026.
8. Patel S, Peng FZ, Fraser K, Gandavadi D, Dwivedy A, Wang X, **Chatterjee P**, Yao S. AptaBLE: a Deep Learning Platform for Aptamer Generation and Analysis. *bioRxiv*, January 2026.
9. Chen T, Quinn Z, Mishra K, O'Connor EC, Silver SE, Zhang Y, Valencia MJ, Mei Y, Behmoaras J, Ferreira LMR†, **Chatterjee P**†. moPPIt: *De Novo* Generation of Motif-Specific and Functionally Active Peptide Binders via Discrete Flow Matching. *bioRxiv*, January 2026.
10. Zhang Y, Tang S, Chen T, Mahood E, Vincoff S, **Chatterjee P**†. PeptiVerse: A Unified Platform for Therapeutic Peptide Property Prediction. *bioRxiv*, January 2026.
11. Tang S, Zhang Y, **Chatterjee P**†. Entangled Schrödinger Bridge Matching. *arXiv*, November 2025.
12. Chen T, Zhang Y, **Chatterjee P**†. AReUReDi: Annealed Rectified Updates for Refining Discrete Flows with Multi-Objective Guidance. *arXiv*, September 2025.
13. Tang S, Zhu Y, Tao M, **Chatterjee P**†. TR2-D2: Tree Search Guided Trajectory-Aware Fine-Tuning for Discrete Diffusion. *arXiv*, September 2025.
14. Goel S, Schray PM, Zhang Y, Vincoff S, Kratochvil H, **Chatterjee P†.** Token-Level Guided Discrete Diffusion for Membrane Protein Design. *arXiv*, September 2025.
15. Zhang Y, Srijay D, Quinn Z, **Chatterjee P**†. Metalorian: De Novo Generation of Heavy Metal-Binding Peptides with Classifier-Guided Diffusion Sampling. *bioRxiv*, July 2025.
16. Tang S, Zhang Y, Tong A, **Chatterjee P**†. Branched Schrödinger Bridge Matching. *arXiv*, June 2025.
17. Chen LT, Pulugurta R, Vure P, **Chatterjee P**†. DPAC: Prediction and Design of Protein-DNA Interactions via Sequence-Based Contrastive Learning. *bioRxiv,* May 2025.
18. Chen T, Zhang Y, Tang S, **Chatterjee P**†. Multi-Objective-Guided Discrete Flow Matching for Controllable Biological Sequence Design. *arXiv*, May 2025.
19. Tang S, Zhang Y, Tong A, **Chatterjee P**†. Gumbel-Softmax Flow Matching with Straight-Through Guidance for Controllable Biological Sequence Generation. *arXiv*, March 2025.
20. Patel S, Peng FZ, Fraser K, **Chatterjee P**†, Yao S†. EVOFLOW-RNA: Generating and representing non-coding RNA with a language model. *bioRxiv*, February 2025.
21. Peng Z, Bezemek Z, Patel S, Yao S, Rector-Brooks J, Tong A†, **Chatterjee P**†. Path Planning for Masked Diffusion Model Sampling. *arXiv*, February 2025.
22. Torres MDT, Chen T, Wan F, **Chatterjee P**†, de la Fuente-Nunez, C†. Generative latent diffusion language modeling yields anti-infective synthetic peptides. *bioRxiv*, February 2025.
23. Tang S, Zhang Y, **Chatterjee P**†. PepTune: De Novo Generation of Therapeutic Peptides with Multi-Objective-Guided Discrete Diffusion. *arXiv*, December 2024.
24. Wang L, Pulugurta R, Vure P, Pal A, Zhang Y, **Chatterjee P**†. PepDoRA: A Unified Peptide Language Model via Weight-Decomposed Low-Rank Adaptation. *arXiv*, October 2024.
25. Goel S, Thoutam V, Marroquin EM, Gokaslan A, Firouzbakht A, Vincoff S, Kuleshov V, Kratochvil HT, **Chatterjee P**†. MeMDLM: De Novo Membrane Protein Design with Masked Discrete Diffusion Protein Language Models. *arXiv*, October 2024.
26. Rector-Brooks J, Hasan M, Peng Z, Quinn Z, Liu C, Mittal S, Dziri N, Bronstein M, Bengio Y, **Chatterjee P**, Tong A, Bose AJ. Steering Masked Discrete Diffusion Models via Discrete Denoising Posterior Prediction. *arXiv*, October 2024.

27. Chen T, Zhang Y, **Chatterjee P†.** moPPIt: *De Novo* Generation of Motif-Specific Binders with Protein Language Models. *bioRxiv*, August 2024.
28. Bhat S, Palepu K, Hong L, Mao J, Ye T, Iyer R, Zhao L, Chen T, Vincoff S, Watson R, Wang T, Srijay D, Kavirayuni VS, Kholina K, Goel S, Vure P, Deshpande AH, Soderling S, DeLisa M, **Chatterjee P†.** *De Novo* Design of Peptide Binders to Conformationally Diverse Targets with Contrastive Language Modeling. *bioRxiv*, July 2024.
29. Zhang Y, He P, Hsu A, **Chatterjee P†**. MetaLATTE: Metal Binding Prediction via Multi-Task Learning on Protein Language Model Latents. *bioRxiv*, 2024.
30. Vincoff S, Goel S, Kholina K, Pulugurta R, Vure P, **Chatterjee P†**. FusOn-pLM: A Fusion Oncoprotein-Specific Language Model via Focused Probabilistic Masking. *bioRxiv*, 2024.
31. Chen T, Vure P, Pulugurta R, **Chatterjee P†**. AMP-Diffusion: Integrating Latent Diffusion with Protein Language Models for Antimicrobial Peptide Generation. *bioRxiv*, 2024.
32. Peng Z, Schussheim B, **Chatterjee P†**. PTM-Mamba: A PTM-Aware Protein Language Model with Bidirectional Gated Mamba Blocks. *bioRxiv*, 2024.
33. Chen T, Pertsemlidis S, Kavirayuni VS, Vure P, Pulugurta R, Hsu A, Vincoff S, Yudistyra V, Hong L, Haarer E, Zhao L, **Chatterjee P†**. PepMLM: Target Sequence-Conditioned Generation of Peptide Binders via Masked Language Modeling. *arXiv*, October 2023.
34. Bhat S, Palepu K, Yudistyra V, Hong L, Kavirayuni VS, Chen T, Zhao L, Wang T, Vincoff S, **Chatterjee P†**. *De Novo* Generation and Prioritization of Target-Binding Peptide Motifs from Sequence Alone. *bioRxiv*, June 2023.
35. Brixi G, Ye T, Palepu K, Hong L, Yudistyra V, Vincoff S, Christopher J, Li X, Bhat S, Monticello C, Lopez-Barbosa N, Petersen L, Zang TL, Liu T, Zhao L, Zhang S, Tysinger E, Stan T, Koseki S, DeLisa M, **Chatterjee P†**. Design of Peptide-Guided Protein Degraders with Structure-Agnostic Language Models. *Research Square*, April 2023.
36. Koseki S, Hong L, Yudistyra V, Stan T, Tysinger E, Silverstein RA, Kramme C, Amrani N, Savic N, Pacesa M, Rodriguez T, Ponnapati M, Jacobson JM, Church G, Truant R, Jinek M, Kleinstiver BP, Sontheimer EJ, **Chatterjee P†.** PAM-Flexible Genome Editing with an Engineered Chimeric Cas9. *Research Square*, February 2023.
37. Koseki S, Hong L, Yudistyra V, Stan T, Tysinger E, Silverstein RA, Kramme C, Amrani N, Savic N, Pacesa M, Rodriguez T, Ponnapati M, Jacobson JM, Church G, Truant R, Jinek M, Kleinstiver BP, Sontheimer EJ, **Chatterjee P†.** PAM-Flexible Genome Editing with an Engineered Chimeric Cas9. *Research Square*, February 2023.
38. Kramme C, Smela MP, Wolf B, Fortuna P, Brixi G, Palepu K, Dong E, Adams J, Bhat S, Koseki S, Tysinger E, Stan T, Kohman R, Liu S, Kobayashi M, Shioda T, Church G, **Chatterjee P†**. Efficient Human Germ Cell Specification from Stem Cells via Combinatorial Expression of Transcription Factors. *bioRxiv*, July 2022.
39. Smela MP, Kramme C, Fortuna P, Adams J, Dong E, Kobayashi M, Brixi G, Tysinger E, Kohman R, Shioda T, **Chatterjee P†**, Church G. Directed Differentiation of Human iPSCs to Functional Ovarian Granulosa-Like Cells via Transcription Factor Overexpression. *bioRxiv*, July 2022.
40. Palepu K, Ponnapati M, Bhat S, Tysinger E, Stan T, Brixi G, Koseki S, **Chatterjee P†**. Design of Peptide-Based Protein Degraders via Contrastive Deep Learning. *bioRxiv*, May 2022.
41. Tripathy SP, Ponnapati M, Jacobson JM, **Chatterjee P†**. Sub-Picomolar Detection of SARS-CoV-2 RBD via Computationally-Optimized Peptide Beacons. *bioRxiv*, June 2021.
42. Costa A, Ponnapati M, Jacobson JM, **Chatterjee P**. Distillation of MSA Embeddings to Folded Protein Structures with Graph Transformers. *bioRxiv*, June 2021.
43. **Chatterjee P†**, Ponnapati M, Jacobson JM. Targeted Intracellular Degradation of SARS-CoV-2 RBD via Computationally-Optimized Peptide Fusions. *bioRxiv*, June 2020.
44. **Chatterjee P†**, Jakimo N, Jacobson JM. Robust Genome Editing of Single-Base PAM Targets with Engineered ScCas9 Variants. *bioRxiv*, April 2019.
45. Jakimo N*, **Chatterjee P***, Nip L*, Jacobson JM. A Cas9 with Complete PAM Recognition for Adenine Dinucleotides. *bioRxiv*, September 2018.
46. Jakimo N*, **Chatterjee P**, Jacobson JM. ssRNA/DNA-Sensors via Embedded Strand-Displacement Programs in CRISPR/Cas9 Guides. *bioRxiv*, February 2018.
47. **Chatterjee P†**, Jakimo N, Jacobson JM. Divergent PAM Specificity of a Highly-Similar SpCas9 Ortholog. *bioRxiv*, February 2017.
48. Jakimo N, **Chatterjee P**, Jacobson JM. Chimeric CRISPR guides enhance Cas9 target specificity. *bioRxiv*, June 2017.

†Corresponding author
***Equal-contribution first author
Google Scholar: https://scholar.google.com/citations?user=XExgv9YAAAAJ&hl=en&oi=ao
ORCID: 0000-0003-3957-8478

Teaching

Instructor and Course Director Fall 2026 –
University of Pennsylvania
CIS 6270: Discrete Generative Models

Instructor and Course Director Spring 2026 –
University of Pennsylvania
BE 3060: Cellular Engineering

Instructor and Course Director Spring 2023 – Spring 2025
Duke University
BME590: Generative AI for Protein Design

Instructor and Course Director Fall 2022 – Fall 2025
Duke University
BME390L: Molecular Engineering

Instructor and Course Director January 2019 and 2020
Massachusetts Institute of Technology
CRISPR: Engineering the Genome

Graduate Teaching Assistant Spring 2019-2020
MIT Media Lab
MAS.S61: How To Grow Almost Anything (HTGAA)

Undergraduate Teaching Assistant Fall 2015
Massachusetts Institute of Technology
6.01: Introduction to EECS

Patents

1. Chatterjee, P., et al. "Methods and Compositions for Producing Granulosa-Like Cells." US Patent Application 19/361,182; published as US20260049287A1, February 19, 2026.
2. Chatterjee, P., et al. "Artificial Intelligence Designed Artificial Immune Receptors Specific for a Synthetic Ligand and Uses Thereof." US Provisional Patent Application 63/955,760, January 6, 2026.
3. Chatterjee, P., et al. "Minimal Peptide Fusions for Targeted Intracellular Degradation of FOXP3." US Patent Application Publication US20250340856A1, November 6, 2025.
4. Chatterjee, P., et al. "Methods and Compositions for Producing Primordial Germ Cell-Like Cells." US Patent Application Publication US20250207091A1, June 26, 2025.
5. Chatterjee, P., et al. "Post-Translational Modification Chimeric Molecule." US Patent Application Publication US20240335554A1, October 10, 2024.
6. Chatterjee, P., et al. "Compositions for the Targeted Degradation of a Protein and Methods of Making and Using Same." US Provisional Patent Application 63/673,939, July 22, 2024.
7. Chatterjee, P., et al. "Compositions for the Targeted Degradation of a Protein and Methods of Making and Using Same." US Provisional Patent Application 63/673,921, July 22, 2024.
8. Chatterjee, P., et al. "Compositions for the Targeted Degradation of a Protein and Methods of Making and Using Same." US Provisional Patent Application 63/673,911, July 22, 2024.
9. Chatterjee, P., et al. "Systems and Methods for Generating Peptide Binders via Masked Language Modeling." US Provisional Patent Application 63/673,900, July 22, 2024.
10. Chatterjee, P., et al. "Peptide-Guided Protein Stabilizers and Methods of Making and Using Same." US Patent Application 18/666,500, May 16, 2024.
11. Chatterjee, P., et al. "Methods and Compositions for Producing Oogonia-Like Cells." International Patent Application Publication WO2023192939A3, November 9, 2023.
12. Chatterjee, P., et al. "Beta-Catenin Protein Degradation." International Patent Application Publication WO2023150788A2, August 10, 2023.
13. Chatterjee, P., et al. "Peptide-Based Probes for the Detection of SARS-CoV-2." US Patent Application Publication US20230194525A1, June 22, 2023.
14. Chatterjee, P., et al. "Peptide-Guided Protein Stabilizers and Uses Thereof." US Provisional Patent Application 63/451,711, March 13, 2023.
15. Chatterjee, P., et al. "Platform and Method for Determining Critical Transcription Factors (TF) for TF-Based Human Induced Pluripotent Stem Cell (hiPSC) Differentiation." US Patent Application 17/981,881, filed November 7, 2022; published as US20230144683A1, May 11, 2023.

16. Chatterjee, P., et al. "Sequence-Based Framework to Design Peptide-Guided Degraders." US Provisional Patent Application 63/423,320, filed November 7, 2022; related European Patent Application Publication EP4616405A2, September 17, 2025.
17. Chatterjee, P., et al. "Compositions and Methods for Gene Editing with Woolly Mammoth Alleles." International Patent Application Publication WO2022125940A2, June 16, 2022.
18. Chatterjee, P., et al. "Applications of Recombined ScCas9 Enzymes for PAM-Free DNA Modification." US Patent Application 17/841,639, filed June 15, 2022; US Patent US12529042B2, granted January 20, 2026.
19. Chatterjee, P., et al. "Distillation of MSA Embeddings to Folded Protein Structures using Graph Transformers." US Patent Application 17/831,435, filed June 2, 2022; published as US20220392566A1, December 8, 2022.
20. Chatterjee, P., et al. "Contrastive Learning for Peptide-Based Degrader Design and Uses Thereof." US Provisional Patent Application 63/344,820, filed May 23, 2022; US Patent Application Publication US20250335663A1, October 30, 2025.
21. Chatterjee, P., et al. "Minimal Peptide Fusions for Targeted Intracellular Protein Degradation." US Patent Application Publication US20220025341A1, January 27, 2022.
22. Chatterjee, P., et al. "Computationally-Optimized ACE2 Peptides for Competitive Interception of SARS-CoV-2." US Patent Application Publication US20220002700A1, January 6, 2022.
23. Chatterjee, P., et al. "Applications of Engineered Streptococcus Canis Cas9 Variants on Single-Base PAM Targets." US Patent Application Publication US20200332271A1, October 22, 2020; US Patents US11453865B2 (September 27, 2022), US11697808B2 (July 11, 2023), and US12385024B2 (August 12, 2025).
24. Chatterjee, P., et al. "Applications of Streptococcus-Derived Cas9 Nucleases on Minimal Adenine-Rich PAM Targets." International Patent Application Publication WO2019217336A2, November 14, 2019.
25. Chatterjee, P., et al. "Streptococcus Canis Cas9 as a Genome Engineering Platform with Novel PAM Specificity." US Patent Application Publication US20190218532A1, July 18, 2019; related US Patent US12054755B2, granted August 6, 2024.
26. Chatterjee, P., et al. "Chimeric DNA:RNA Guide for High Accuracy Cas9 Genome Editing." US Patent Application Publication US20180282722A1, October 4, 2018.

Scientific Honors and Awards

- Hartwell Biomedical Research Collaboration Award 2026
- Maximizing Investigators Research Award, NIH/NIGMS 2024
- Duke-Coulter Partnership Award, Wallace H. Coulter Foundation 2023
- Hartwell Individual Biomedical Research Award 2023
- Guest Editor, Current Opinions in Biomedical Engineering 2022-2023
- Carlos M. Varsavsky Research Fellowship 2020-2022
- Top Young Scientist Award – Frontiers in Genome Engineering, Kobe, Japan 2019
- SuperUROP Top Undergraduate Thesis Award 2016
- Top Undergraduate Abstract – American Society of Hematology 2012
- Outstanding Achievement Award – American Society of Hematology 2012, 2013
- Harvard Summer Immunology Research Fellowship 2012, 2013
- Lemelson-MIT InvenTeams Grant Recipient 2010

Invited Talks

- St. Jude Children's Research Hospital - Structural & Computational Biology Symposium 2027
- New England Biolabs Seminar 2027
- Icahn School of Medicine at Mount Sinai - Microbiology Seminar 2027
- PEGS Boston - Emerging Peptide Therapeutics 2027
- ACS Spring Meeting - BIOT-YCC Symposium: Preparing the Next Generation of Biotechnology Leaders 2027
- AI in Drug Discovery Conference, London 2027
- Duke University - Computational Biology and Bioinformatics Seminar 2027
- NASSCOM Technology & Leadership Forum, Mumbai 2027
- St. Jude Children's Research Hospital - Computational Biology Seminar 2027
- Broad Institute - Models, Inference & Algorithms (MIA) Seminar 2027
- PEGS AI - Computational Models for Advancing Biologics 2027
- University of Chicago - Pritzker School of Molecular Engineering Seminar 2027
- Peptide Therapeutics MicroSummit - Recorded Presentation 2026
- **Keynote Speaker** - University of Florida Health Cancer Institute AI Symposium 2026
- PEGS Europe Invited Speaker 2026
- University of Utah - AI Interest Group Seminar 2026
- Discovery On Target Invited Speaker 2026
- Instituto de Biotecnología, UNAM Seminar 2026

- **Keynote Speaker** - Lewis Katz School of Medicine (Temple University) Research Symposium 2026
- Penn HTSC and HIT-ID Symposium Invited Speaker 2026
- ACS Webinar - AI in Drug Discovery 2026
- AskBio Seminar 2026
- Genetic Engineering Technology Days (CIA) 2026
- Penn Microbiology Seminar 2026
- City University of New York (CUNY) Seminar 2026
- **Keynote Speaker** - StartART Congress 2026
- Genentech Seminar 2026
- Generative AI Lecture: Discovery on Target Symposium 2026
- Mammalian Synthetic Biology Workshop 2026
- **State of the Art Lecture** – American Society of Virology Annual Meeting 2026
- Aktis AI Seminar 2026
- **Plenary Speaker** – Spring School in Synthetic Biology at Scuola Normale Superiore 2026
- **Keynote** - Pacific Coast Reproductive Society Annual Meeting 2026
- OPT Peptides & Emerging Drug Conjugates 2026
- NanoBio Seminar, MIT Media Lab 2026
- How to Grow Almost Anything Seminar, MIT Media Lab 2026
- Chemical and Biomolecular Engineering Seminar - Cornell University 2026
- **Keynote** – 34th Nasscom Technology & Leadership Forum 2026
- Gordon Research Conference on Protein Folding and Dynamics 2026
- SickKids Toronto – Molecular Medicine Seminar 2025
- Sanford Burnham Prebys Center for Data Sciences and AI Seminar 2025
- SimBioChem Workshop – EurIPS Conference 2025
- AI4Science Seminar Series - Georgia Institute of Technology 2025
- Penn ASSET Seminar 2025
- Genetic Agency Technology Conference 2025
- Carolina Biophysics Symposium 2025
- **Keynote** - IX Congreso de Fisicoquímica, Estructura y Diseño de Proteínas 2025
- **Keynote** - Penn Systems Pharmacology and Translational Therapeutics Retreat 2025
- ASSET Center Seminar – University of Pennsylvania 2025
- CECAM Generative AI and Large Language Models for Protein Modeling 2025
- Stanford AI Lab Seminar 2025
- Machine Learning for Protein Engineering (ML4PE) Seminar 2025
- Prague bioML Symposium, IOCB Prague 2025
- Seminar at Cradle Bio, Inc. 2025
- Gordon Research Conference on Enzymes, Coenzymes and Metabolic Pathways 2025
- Genomic Instability Think Tank - University of Pennsylvania 2025
- Life Edit Therapeutics, Seminar 2025
- Seminar at National Institute on Drug Abuse (NIH/NIDA) 2025
- Seminar at École Polytechnique Fédérale de Lausanne, Institute of Bioengineering 2025
- **Keynote** - Council of Physicians & Scientists - US Fertility 2025
- International Conference of Learned Representations - AI4NA Workshop 2025
- **Keynote** - Duke-NUS Health Data Science Symposium 2025
- **Keynote** - American Society for Biochemistry and Molecular Biology - Special Interest Group 2025
- DCI Cancer Biology Seminar – From Discovery to Impact 2025
- TriPoD Seminar, University of North Carolina, Chapel Hill 2025
- How to Grow Almost Anything Seminar, MIT Media Lab 2025
- Charting New Horizons for Discovery & Translational Science Symposium, Duke 2025
- University of Pennsylvania, Computer and Information Science Seminar 2025
- Tokyo University of Agriculture and Technology Seminar 2025
- University of Texas, Computer Science Seminar 2024
- 5th Molecular Machine Learning (MoML) Conference 2024
- Harvard Medical School Seminar 2024
- Duke Summit on AI for Health Innovation 2024
- Triangle iGEM Symposium 2024
- Duke Impact of Genomic Variation on Function Consortium 2024
- Duke Center for Computational and Digital Health Innovation 2024
- Center for Precision Health, University of Pennsylvania 2024
- **Keynote** - Canadian Fertility & Andrology Society Annual Meeting 2024

- ICML 2024 Workshop on Accessible and Efficient Foundation Models for Biological Discovery 2024
- University of Massachusetts Medical School, RNA Therapeutics Institute 2024
- Keystone Symposia: Proximity Induced Therapeutics 2024
- Mid-Atlantic Synthetic Biology Network 2024
- University of Pennsylvania, Department of Chemical and Biomolecular Engineering 2023
- Biomedical Engineering Society Meeting 2023
- International Society of Molecular Biology Meeting 2023
- Keystone Symposia: Computational Design of Biomolecules 2023
- Mila, Quebec AI Institute 2022
- Keystone Symposia: Targeted Protein Degradation 2022
- Germ Cells at Cold Spring Harbor Labs 2022
- **Keynote** - NIH/NIEHS Branch Conference 2022
- **Keynote** - Biochemistry and Biomedical Sciences Research Symposium, McMaster University 2022
- Keystone Symposia: Computational Design & Biomolecules 2022
- Keynote Speaker: CRI-Paris Molecular Biology Symposium 2021
- Frontiers of CRISPR/Cas at Cold Spring Harbor Labs 2019
- Frontiers in Genome Engineering Conference, Kobe, Japan 2019

Academic Mentorship

Current Trainees

Postdoctoral Fellows:

- Dr. Yinuo Zhang, Postdoctoral Fellow, University of Pennsylvania 2026-present
- Dr. Ryan Chow, Medical Fellow, University of Pennsylvania 2026-present
- Dr. Elizabeth Mahood, Postdoctoral Fellow, University of Pennsylvania 2025-present
- Dr. Lin Zhao, Postdoctoral Fellow, University of Pennsylvania 2022-present

PhD Students:

- Megan Taylor, PhD Student, University of Pennsylvania 2026-present
- Ruoxi Zhang, PhD Student, University of Pennsylvania 2026-present
- Aastha Pal, PhD Student, University of Pennsylvania 2025-present
- Tong Chen, PhD Student, University of Pennsylvania 2025-present
- Tian Wang, PhD Student, University of Pennsylvania 2024-present
- Sophia Vincoff, PhD Student, University of Pennsylvania 2023-present
- Lauren Hong, PhD Student, University of Pennsylvania 2022-present

Masters Students:

- Sumi Kimura, Masters Student, University of Pennsylvania 2025-present

Undergraduate Students:

- Faraz Rahman, Undergraduate Student, University of Pennsylvania 2026-present
- Navya Nori, Undergraduate Student, Georgia Tech 2026-present
- Bobo Li, Undergraduate Student, University of Pennsylvania 2026-present
- Julia Vizza, Undergraduate Student, Cornell University 2026-present
- Fiona Yan, Undergraduate Student, University of Pennsylvania 2026-present
- Sophia Tang, Undergraduate Student, University of Pennsylvania (CRA Award Winner) 2024-present
- Divya Srijay, Undergraduate Student, Duke University 2023-present
- Shrey Goel, Undergraduate Student, Duke University 2023-present

Former Trainees

PhD Students

- Zachary Quinn, PhD Student, Duke University 2024-2026

Masters Students

- Jiale Chen, Masters Student, University of Pennsylvania 2025-2026
- Yesol Kim, Masters Student, University of Pennsylvania 2025-2026
- Howard Liu, Masters Student, Duke University 2024-2026
- Zhangzhi (Fred) Peng, PhD Student, Duke University 2023-2025
- Tianlai Chen, Masters Student , Duke University 2022-2025

Undergraduate Students

- Kimberly Liang, Undergraduate Student, University of Pennsylvania 2025-2026
- Ananya Krishna, Undergraduate Student, Yale University 2024-2025
- Vatsal Parikh, Undergraduate Student, Princeton University 2024-2025
- Ishaan Mehrotra, Undergraduate Student, Duke University 2024-2025
- Shubhika Munot, Undergraduate Student, Duke University 2024-2025
- Kishan Patel, Undergraduate Student, Duke University 2024-2025
- Pranay Vure, Undergraduate Student, Duke University (NSF GRFP Recipient) 2023-2025
- Kseniia Kholina, Undergraduate Student, Duke University 2023-2025
- Rishab Pulugurta, Undergraduate Student 2023-2025
- Srikar Kavirayuni, Undergraduate Student, Duke University 2022-2025
- Leyao Wang, Undergraduate Student, Vanderbilt University 2024-2025
- Phil He, Undergraduate Student, Duke University 2023-2024
- Samuel Scheinbach, Undergraduate Student, University of Rochester 2024
- Benjamin Schussheim, Undergraduate Student, Duke University 2023-2024
- Ashley Hsu, Undergraduate Student, Duke University 2023-2024
- Suhaas Bhat, Undergraduate Student, Harvard University (Rhodes Scholar) 2021-2024
- Garyk Brixi, Undergraduate Student, Harvard University (NSF GRFP Recipient) 2022-2023
- Kalyan Palepu, Undergraduate Student, Harvard University 2020-2023
- Amirabbas Kazeminia, Undergraduate Student, MIT 2022-2023
- Teodora Stan, Undergraduate Student, MIT 2021-2022
- Emma Tysinger, Undergraduate Student, MIT 2020-2022
- Sabrina Koseki, Undergraduate Student, Wellesley College (NSF GRFP Recipient) 2019-2022
- Andrew Hennes, Undergraduate Student, MIT 2019
- Megi Topalli, Undergraduate Student, MIT 2019
- Rachel Honigsberg, Undergraduate Student, Boston University 2018

High School Students

- Vishrut Thoutam, Current Undergraduate Student, Princeton University 2024-2025

Outreach

Instructor and Course Director 2026-present
Penn ESAP Program

Education Workshop Host and Organizer 2025-present
Penn GEMS Program

Education Workshop Host and Organizer 2024-2025
Duke AI Day

Research Advisor 2023-2025
DukeREP Program

Middle and High School Instructor 2015-2021
MIT Educational Studies Program

Community Science Director and Robotics Coach 2012-2016
Margaret Fuller Neighborhood House, Cambridge, MA

Interfaith Fellow and Program Director 2012-2015
MIT Addir Fellows Interfaith Dialogue

Independently Acquired Grant Funding

Gift Funding 2026-
Donations from philanthropists and industry
Current Amount: $95,000

Futures Fund 2026-2027
Project Title: Generative Peptides for Precision Health and Sustainability
Current Amount: $100,000

NIH/NHGRI 2026-2030
Project Title: Multigenic Control of Gene Expression in Health and Disease
Current Amount: $600,000

Colton Discovery Pilot Award (Co-PI) 2026-2027
Project Title: AI-guided development of novel therapeutics for rare inflammatory autoimmune disease
Funding Amount: $150,000 (receive $50,000 subaward)

University of Pennsylvania - HIT-ID 2025-2026
Project Title: MoTSelTOP: A Generative AI Platform for Selective Peptide Antagonist Discovery Targeting CNS GPCRs
Funding Amount: $100,000

NIH/NIGMS 2024-2029
Project Title: Towards the design of programmable, isoform-selective proteome editing system
Grant 1R35GM155282-01
Funding Amount: $2,012,500

EndAxD 2023-2025
Project Title: Design of a GFAP degradation platform for the treatment of Alexander's Disease
Funding Amount: $50,000

NIH/NCI 2023-2025
Project Title: Programmable peptide-guided protein degradation
Grant 1R21-CA278468-01A1
Funding Amount: $414,655

NIH/NCI 2023-2025
Project Title: Proteasomal recruiters of PAX3-FOXO1 Designed via Sequence-Based Generative Models
Supplement to Grant 3U54-CA231630-01A1S4 (PD/PI: Corinne Linardic)
Funding Amount: $157,719

Homeworld Collective 2023-2024
Project Title: Generative Design of Programmable Metal-Binding Proteins for Bioremediation
Funding Amount: $82,500

CHDI Foundation 2023-2025
Project Title: Design of mHTT and MSH3 Degraders with Language Model-Derived Peptide Guides
Funding Amount: $610,504

Wallace H. Coulter Foundation 2023-2024
Project Title: Selective Degradation of PAX3-FOXO1 with Language Model-Derived Peptide Guides
Funding Amount: $281,963

The Hartwell Foundation 2023-2026
Project Title: Design of a TRIM8 degradation platform for the treatment of Ewing sarcoma
Funding Amount: $300,000

Duke Startup Funding 2022-2025
Funding Amount: $1.4 million

Commercial Activities

Co-Founder and Scientific Advisor 2026-present
Kosha Sciences, Inc.
- Preseed startup to develop multi-objective-guided generative models for lipid nanoparticle design

Co-Founder and Scientific Advisor 2025-present
Recognition Bio, Inc.
- Preseed startup to develop multi-objective-guided generative models for peptide-drug conjugate design

Co-Founder and Scientific Advisor 2024-present
Atom Bioworks, Inc.
- Seed-funded startup to develop generative language models for RNA aptamer design and mRNA optimization

Co-Founder and Scientific Advisory Board Member 2020-present
Gameto, Inc.
- Series B-funded biotechnology company to develop fertility solutions from patented ovarian organoid technology

Co-Founder and Scientific Advisor 2020-present
UbiquiTx, Inc.
- Seed-funded startup to develop ubiquibody technology for intracellular degradation of pathogenic targets

Scientific Advisory Board Member 2025-present
Output Biosciences, Inc.

Scientific Advisory Board Member 2025-present
Rime Therapeutics, Inc.

Scientific Advisory Board Member 2025-present
Latus Bio, Inc.

Scientific Advisory Board Member 2024-present
Juniper Genomics, Inc.

Scientific Advisory Board Member 2024-present
Infinity Biosciences, Inc.

Scientific Advisory Board Member 2023-present
DreamFold, Inc.

Scientific Advisor 2023-2025
Form Bio, Inc.

Scientific Advisor 2020-2025
Yokogawa Electric Corporation

Professional Societies and Services

Workshop Program Committee 2026-present
International Conference on Learning Representations (ICLR)

Editorial Board Member 2026-present
Current Opinion in Chemical Biology

Ad-Hoc Study Section Member 2026-present
National Institutes of Health
Macromolecular Structure and Function A (MSFA) Study Section

Ad-Hoc Study Section Member 2026-present
National Institutes of Health
Macromolecular Structure and Function B (MSFB) Study Section

Active Member 2025-present
Association for the Advancement of Artificial Intelligence (AAAI)

Workshop Organizer 2026
Neural Information Processing Systems (NeurIPS)
BeNTo: Beyond Next Token Prediction

Workshop Organizer 2025, 2026
Neural Information Processing Systems (NeurIPS)
SimBioChem: Machine Learning for Simulations in Biology and Chemistry

Workshop Organizer 2023, 2024, 2025, 2026
International Conference on Learning Representations (ICLR)
GEM: Generative and Experimental Perspectives for Biomolecular Design Workshop

Workshop Steering Committee Board Member 2023
International Conference on Machine Learning (ICML)
Machine Learning for Life and Materials Science Workshop